

RESEARCH SUMMARY — MACHINE LEARNING

Abstract

This paper summarizes an experiment comparing two supervised learning approaches for binary classification.

Methodology

The dataset was divided into training, validation, and test sets. Models were evaluated using accuracy, precision, recall, F1 score, and ROC-AUC.

Results

The nonlinear model achieved stronger recall while the linear baseline produced fewer false positives.

Conclusion

Model selection should consider the operational cost of false positives and false negatives rather than accuracy alone.